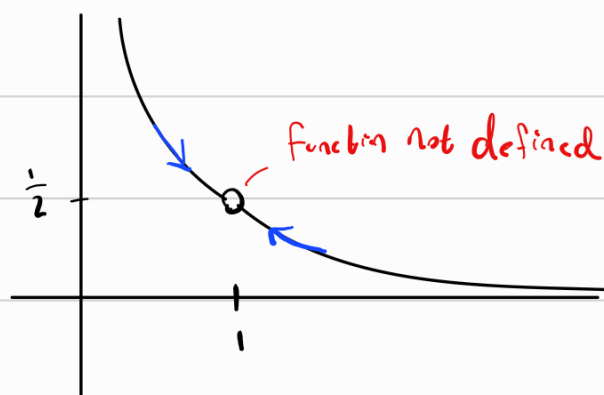


Limits Numerically and Graphically

A limit addresses this idea of 'closeness' for functions. That is, we want to look at what happens to a function as the input values get extremely close to a given value. In fact, a limit can exist even when a function isn't defined at a point.

Consider $f(x) = \frac{x-1}{x^2-1}$



$f(x)$ isn't defined at $x=1$, but the limit as $x \rightarrow 1$ is $\frac{1}{2}$ since the function still gets very close to $\frac{1}{2}$ as x gets close to 1. We write

$$\lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = \frac{1}{2}$$

we'll have a more precise definition later

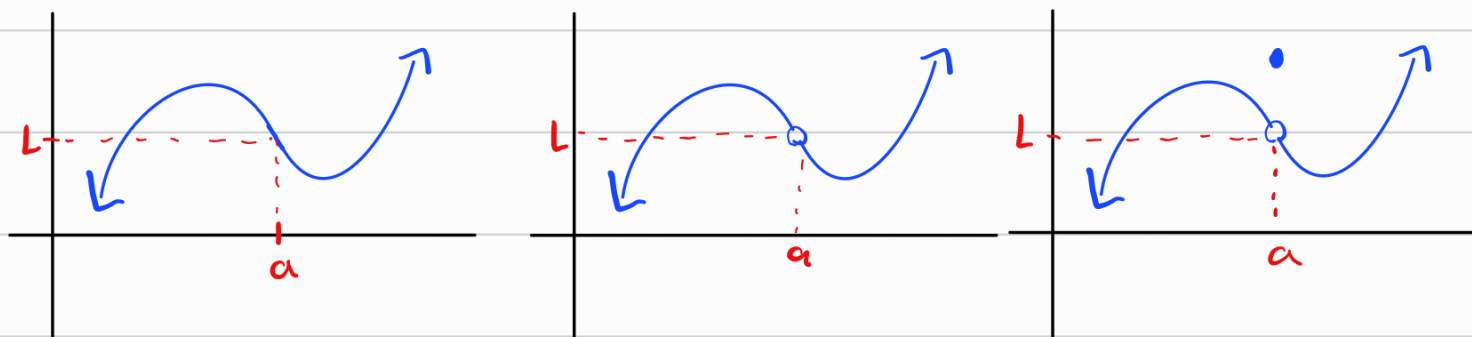
Def:

Suppose $f(x)$ is defined near a , but not necessarily at a , we write

$$\lim_{x \rightarrow a} f(x) = L \quad \text{'limit of } f(x) \text{ as } x \text{ approaches } a, \text{ equals } L'$$

if the values of $f(x)$ become arbitrarily close to L as x gets close to a (but not equal to a).

Notice how this definition doesn't need $f(x)$ to be defined.



in all three graphs, $\lim_{x \rightarrow a} f(x) = L$.

Sometimes, rarely, we check numerically by evaluating the function at values closer to a .

Ex: $f(x) = \frac{\sin(x)}{x}$, $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$

Note, $f(x)$ is
even so
 $f(-1) = f(1)$

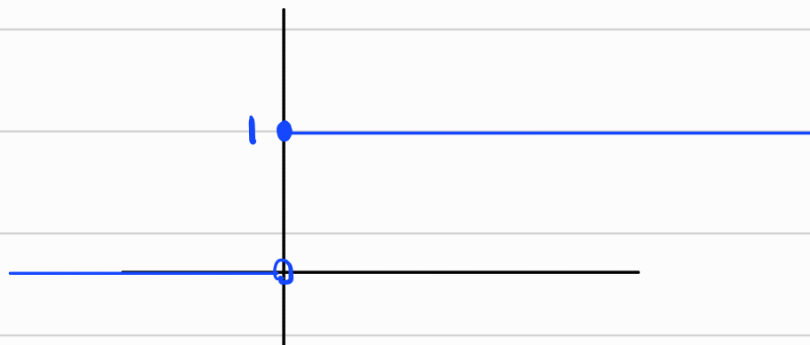
x	$f(x)$
± 1	0.84147
± 0.5	0.95885
± 0.1	0.99833
± 0.01	0.99998
± 0.001	0.99999

Approximation

Sided limits

As a motivating example, consider the Heaviside function

$$H(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$



as $t \rightarrow 0$, the function doesn't seem to go to any one value. On one side, it goes to 0, on the other it goes to 1. For situations like this, we use sided limits.

Def:

We write $\lim_{x \rightarrow a^-} f(x) = L$ if as x approaches a while $x < a$ (always) then $f(x) \rightarrow L$.

We write $\lim_{x \rightarrow a^+} f(x) = L$ if as x approaches a while $x > a$ (always) then $f(x) \rightarrow L$.

In the case that the left and right limit disagree, we say the $\lim_{x \rightarrow a} f(x)$ does not exist.

In fact,

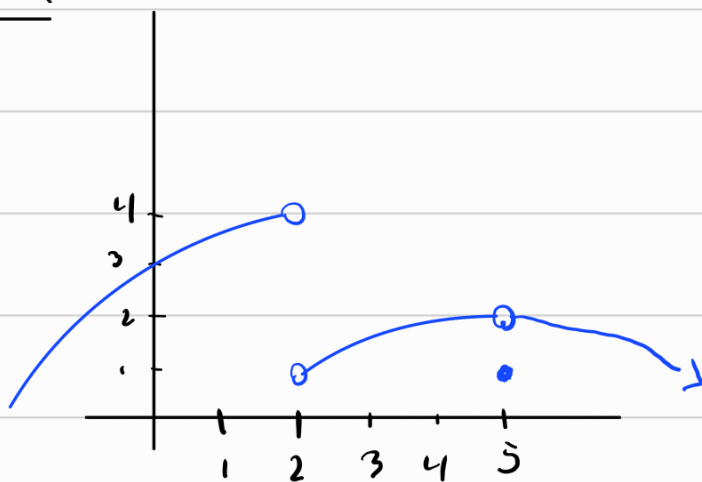
$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

rather

$$\text{If } \lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x) \text{ then } \lim_{x \rightarrow a} f(x) \text{ DNE}$$

does not
exist

Example:



$$\lim_{x \rightarrow 2^-} f(x) = 4$$

$$\lim_{x \rightarrow 5^-} f(x) = 2$$

$$\lim_{x \rightarrow 2^+} f(x) = 1$$

$$\lim_{x \rightarrow 5^+} f(x) = 2$$

$$\lim_{x \rightarrow 2} f(x) = \text{DNE}$$

$$\lim_{x \rightarrow 5} f(x) = 2$$

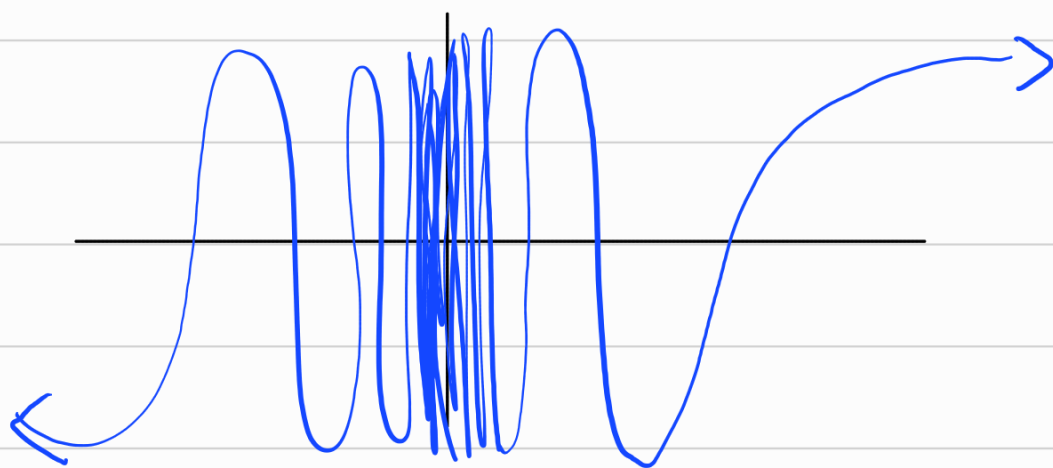
$$f(2) = \text{undefined}$$

$$f(5) = 1$$

Another way limits don't exist

We see that if the left and right disagree, then the entire limit won't exist. Here we investigate another way they fail to exist.

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right) \text{ DNE}$$



The values close to 0 jump up and down between ± 1 an infinite number of times. In fact, we can find a sequence of numbers st.

$$\sin\left(\frac{\pi}{x}\right) \rightarrow 1 \quad \text{if} \quad x = 2, \frac{2}{5}, \frac{2}{9}, \frac{2}{13}, \dots, \frac{2}{4n+1} \rightarrow 0$$

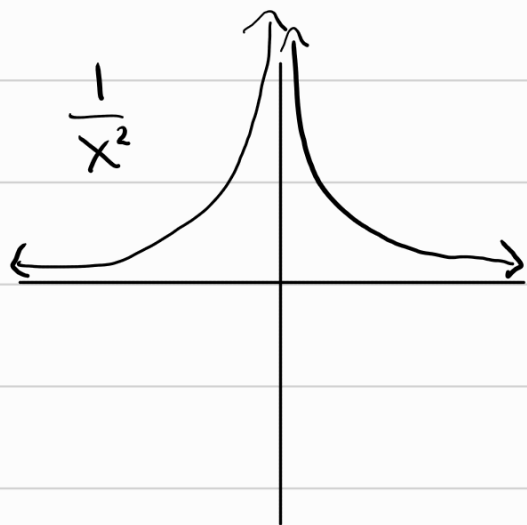
$$\sin\left(\frac{\pi}{x}\right) \rightarrow 0 \quad \text{if} \quad x = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n} \rightarrow 0$$

$$\sin\left(\frac{\pi}{x}\right) \rightarrow -1 \quad \text{if} \quad x = \frac{2}{3}, \frac{2}{7}, \frac{2}{11}, \dots, \frac{2}{4n+3} \rightarrow 0$$

So there are always ways for $x \rightarrow 0$ where $\sin(\frac{\pi}{x})$ can go to any number between -1 and 1. So $\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right) \text{ DNE}$.

Vertical Asymptotes

These are very common in functions. Consider



as x gets close to 0,
 $f(x)$ gets extremely large.

In cases like this, we write

$$\lim_{x \rightarrow a} f(x) = \infty$$

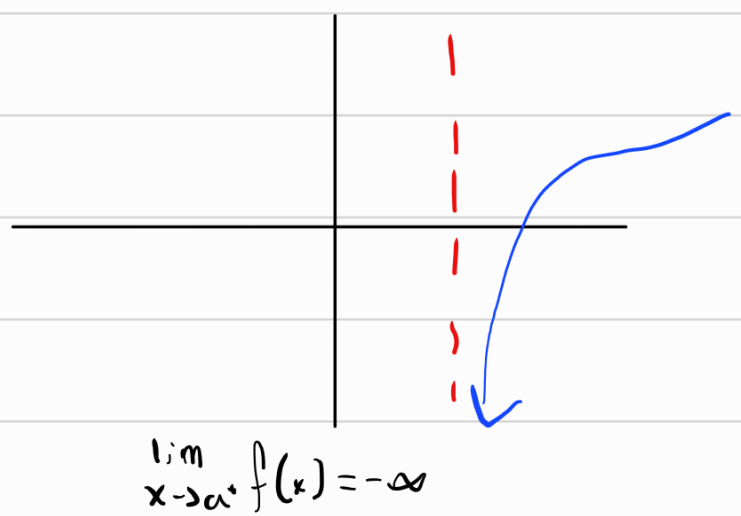
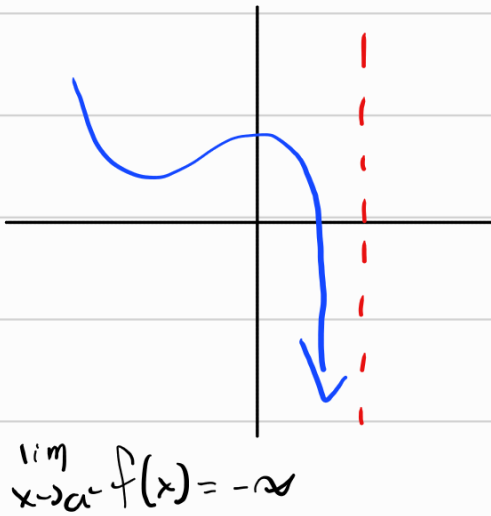
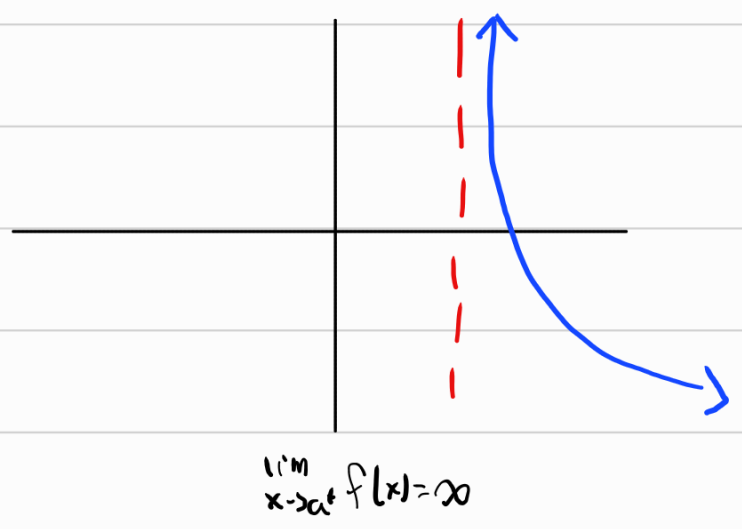
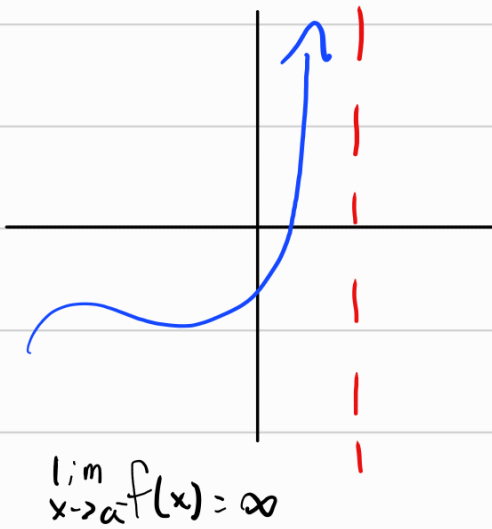
Def:

Let f be a function defined around a , except possibly not at a , then

$$\lim_{x \rightarrow a} f(x) = \infty \quad (\text{or } -\infty)$$

means the values of $f(x)$ become arbitrarily large (or negative) as x approaches a .

Note, the same applies to sided limits



Def:

The vertical line $x=a$ is called a vertical asymptote of $f(x)$ if at least one of the following is true.

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

Note: Just one of these need to be satisfied. Consider $f(x) = \frac{1}{x}$ or $f(x) = \tan(x)$